

Experimental Setup

Does the frozen radial Test 6 clock pass the response prerequisite before Test 7?

Experiment: gross.clock-response.v1 Model: signal-space.ss-ocf-1.flat-neutral-clock.v1
Runtime configuration: fixtures/research/gross-test-06-response.json
Plan lock: 844b2b65a4015533d6b28dcf7941ad493dec84ccb1060b44a2f2dea7fff2a4bd

MODEL AND NUMERICAL SETUP

Action: SS OCF 1 flat action, $U=s-s^2+s^3$, $V=0.25-0.4s+0.2s^2$, $\zeta=0.1$. Neutral response extension $Z=1+0.2s$ ($M_0=m=1$); same underlying action, separate registered active-neutral sector.

Units: $c=\hbar=m=1$; r,t in m^{-1} ; fields in m ; frequencies in m ; dimensionless tick shift

$\Omega\delta_t/(2\pi)$.

Domain: Saved $\omega_Q=0.900$ profiles; $R=80, h=0.1$ and $h=0.05$. Response $R=160$ independently. Fixed readout $r=0.1$.

Boundaries: Regular origin and zero radial fields at finite R . Longevity uses original logged sponge beyond $r=60$. Response uses no sponge; earliest outer reflection cannot reach core by $t=100$.

Initial Data: Exact saved discrete profiles/eigenmodes; $\chi=0.001 f/\max(f/r)$, χ momentum=0, Φ momentum= $-i*0.9*\Phi$. Response $ra=A*c*\cos^4(\pi*(r-c)/(2*w))$ inside $|r-c|<w$, zero outside; $(c,w)=(18,3)$ incoming, optional (4,2) outgoing. Initial $d_t(ra)=d_r(ra)$ incoming and negative outgoing; $p_a=Z*d_t(ra)$. Amplitude A is the coefficient of this prescribed radial packet; it is not an exact peak a normalization.

Solver: Longevity reuses registered kick-drift-kick at $dt=.04,.02$; all 100 nominal periods, sample every .2. Response uses RK4 of discrete reciprocal Hamiltonian at $dt=.02$, mesh/time .01; samples .1; $t=100$.

First solve exact Taylor coefficients $a_1, \delta\Phi^2, \delta\chi^2$ on evolving no-pulse baseline; save all predictions before full nonlinear runs. No parameter fit. Neutral stiffness $G^T Z_{\text{faces}} G$, G

$b=r_{\text{face}}*d_r(b/r)$.

Tolerances: Locked criteria below. L2 means Euclidean norm of uniformly sampled local traces over $t=0..100$. Numerical error is direct coarse/fine difference, no assumed Richardson order. Deterministic error floor 0.2% for response approximation; unresolved if signal RMS $\leq 1e-12$. Resource ceiling 1800 s wall, 1024 MiB, 100 MiB output.

COMPETING PREDICTIONS

usable: The frozen calibration survives full refined longevity and predicts an even, amplitude-squared local response within registered error controls.

inadequate: Unstable clock, sign-odd response, persistent prediction residual or ambiguous local ticks invalidates the tested approximation.

CONTROLS

frozen: exclude branch or calibration retuning; method: verify saved profile/mode hashes; same physical local radius $r=0.1$ on every grid

numerics: distinguish physical response from discretization and boundaries; method: independent $h/2$ at

Experimental Setup

R=80, dt/2 at h=0.1; response additionally R=160; no change to input physical profile branch
pulse-controls: test the action-derived quadratic forcing; method: matched no pulse; +/-0.004; amplitudes 0.002,0.004,0.008; inward plus outward concentric packet and its global sign reverse

PREREGISTERED DECISIONS

frozen-inputs (frozen-inputs)

Pass: All input SHA-256 values match frozen Test 6 profiles and eigenmodes.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

prediction-first (prediction-first)

Pass: Every saved prediction hash and prediction-lock event precedes every receiver evolution completion, with no nonlinear evolution before lock in registered worker.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

sign-even (sign-even)

Pass: L2 sign-odd clock difference / sign-even difference $<1e-5$ for each grid, using +/-0.004.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

amplitude-square (amplitude-square)

Pass: Normalized clock responses at amplitudes 0.002,0.004,0.008 differ by $<5\%$; weak log slope within 0.1 of 2.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

response-match (response-match)

Pass: L2 single-packet prediction discrepancy $<5\%$ and <5 times (mesh + time + box relative response error + 0.002 declared response-error floor).

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Experimental Setup

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

counter-match (counter-match)

Pass: Same frozen discrepancy criteria on concentric inward/outward packet control at amplitude 0.004.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

common-markers (common-markers)

Pass: Each preparation has ≥ 5 positive-going ticks, same count and ordered uniquely paired ticks with absolute shift $< \text{quarter frozen period}$; predicted tick shifts within $\max(10\% \text{ measured shift}, 1e-5 \text{ cycles})$.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

numerical-controls (numerical-controls)

Pass: Mesh, time and box relative L2 clock-difference errors each $< 5\%$; signal RMS $> 1e-12$ m or unresolved.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

conservation (conservation)

Pass: Maximum charge relative error $< 1e-5$ and matched total-energy residual / initial incident energy $< 1\%$ across grids; discrete Hamiltonian, no sponge in $t \leq 100$.

Fail: Finite, resolved data violates the preregistered inequality.

Unresolved: Missing, nonfinite or under-resolved signal/marker or required control.

Uncertainty: Direct independent mesh, time, box differences; finite sampling; no statistical confidence claim.

VISUAL QUESTIONS

packet-history [evidentiary]: Where do the compact neutral packets travel relative to the clock core?

View: PNG/SVG/PDF and exact CSV; observables: local clock, neutral waveform, control residuals

usable: matched prediction, parity, scaling and persistent ticks

inadequate: resolved failure or missing ticks

Controls: frozen, numerics, pulse-controls; uncertainty: Independent numerical controls; spherical restriction

Experimental Setup

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Saved local samples; matched no-pulse subtraction; explicitly labeled amplitude normalization; no smoothing.

upstream-waveform [evidentiary]: What neutral waveform actually reaches the upstream recording sphere?

View: PNG/SVG/PDF and exact CSV; observables: local clock, neutral waveform, control residuals

usable: matched prediction, parity, scaling and persistent ticks

inadequate: resolved failure or missing ticks

Controls: frozen, numerics, pulse-controls; uncertainty: Independent numerical controls; spherical restriction

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Saved local samples; matched no-pulse subtraction; explicitly labeled amplitude normalization; no smoothing.

prediction-response [evidentiary]: Does the precomputed second-order response predict the local clock difference?

View: PNG/SVG/PDF and exact CSV; observables: local clock, neutral waveform, control residuals

usable: matched prediction, parity, scaling and persistent ticks

inadequate: resolved failure or missing ticks

Controls: frozen, numerics, pulse-controls; uncertainty: Independent numerical controls; spherical restriction

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Saved local samples; matched no-pulse subtraction; explicitly labeled amplitude normalization; no smoothing.

parity-scaling [evidentiary]: Is the clock difference even in packet sign and quadratic in amplitude?

View: PNG/SVG/PDF and exact CSV; observables: local clock, neutral waveform, control residuals

usable: matched prediction, parity, scaling and persistent ticks

inadequate: resolved failure or missing ticks

Controls: frozen, numerics, pulse-controls; uncertainty: Independent numerical controls; spherical restriction

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Saved local samples; matched no-pulse subtraction; explicitly labeled amplitude normalization; no smoothing.

marker-convergence [evidentiary]: Are corresponding local tick shifts reproducible under mesh, time and box controls?

View: PNG/SVG/PDF and exact CSV; observables: local clock, neutral waveform, control residuals

usable: matched prediction, parity, scaling and persistent ticks

inadequate: resolved failure or missing ticks

Controls: frozen, numerics, pulse-controls; uncertainty: Independent numerical controls; spherical restriction

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Saved local samples; matched no-pulse subtraction; explicitly labeled amplitude normalization; no smoothing.

Experimental Setup

RESOURCE LIMITS

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{"max_memory_mb": 1024, "max_output_mb": 100, "max_wall_seconds": 1800}
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PROHIBITED INFERENCES

No emergent spacetime, gravity, protected knot or nonspherical stability claim.
Concentric counterpropagating shells are not opposing beams or a recoil calculation.
This ideal local field readout is not a fully invariant two-object Test 7/9 protocol.